

Carry Save Adder (CSA)

1 Introduction

A **Carry Save Adder (CSA)** is a digital adder architecture designed to add three or more binary numbers simultaneously without propagating carries during the initial addition stages. This technique reduces delay and improves performance, particularly in arithmetic-intensive operations such as:

- Multipliers (e.g., Wallace Tree, Dadda Tree)
- MAC (Multiply-Accumulate) Units
- Digital Signal Processing (DSP) hardware

2 Motivation

Traditional binary adders like Ripple Carry Adders (RCAs) propagate carries from one bit to the next, resulting in increased delay. In contrast, CSA avoids this by storing the carry bits and postponing their addition, enabling faster multi-operand arithmetic.

3 Concept

When adding three n -bit binary numbers: A , B , and C , the CSA generates two intermediate vectors:

- **Sum vector (S)**: Sum bits without considering carry-in from previous positions
- **Carry vector (C)**: Carry bits saved and **shifted left by one bit**

These intermediate vectors are later summed using a conventional adder (e.g., Ripple Carry or CLA) to get the final result.

4 Working Principle

Given 3 input numbers A , B , and C , for each bit position i :

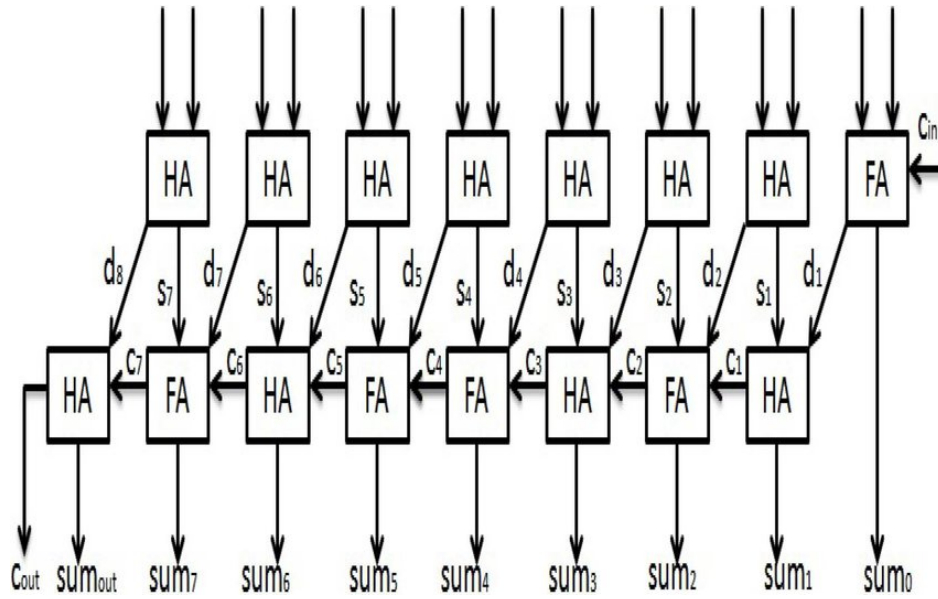
$$S_i = A_i \oplus B_i \oplus C_i$$

$$C_{i+1} = A_i B_i + B_i C_i + A_i C_i$$

- The sum S_i is calculated without considering carry-in from previous stages.
- The carry C_{i+1} is saved for the next position, rather than propagated immediately.

The carry vector C is effectively shifted one bit to the left before final addition.

5 CSA Block Diagram (Conceptual)



6 Numerical Example

Given:

$$A = 1010, \quad B = 1100, \quad C = 0110$$

6.1 Step 1: CSA Output

$$S = A \oplus B \oplus C = 0000$$

$$C = \text{Carry} = 1110 \quad (\text{before shifting})$$

6.2 Step 2: Final Addition

$$\text{Shifted Carry} = C \ll 1 = 11100$$

$$\text{Final Result} = S + (C \ll 1) = 0000 + 11100 = \boxed{11100}$$

7 Advantages

- Eliminates ripple carry delay during intermediate additions
- Enables parallel computation of sum and carry bits
- Ideal for multi-operand addition in fast multipliers

8 Disadvantages

- Requires an additional adder at the end to compute final sum
- More hardware complexity than basic adders

9 Applications

- **Multipliers:** Wallace Tree, Dadda Tree structures use CSA arrays
- **Floating-Point Units:** High-speed mantissa addition
- **Cryptography:** Used in hardware for RSA, AES implementations
- **DSP Hardware:** Efficient implementation of MAC operations

10 Summary

- The Carry Save Adder is an efficient solution for multi-operand binary addition.
- It avoids immediate carry propagation by saving carry bits separately.
- A final stage adder is required to compute the actual binary sum.
- It is essential in high-speed arithmetic architectures.